Reg No.:_____ Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S3 (R.S) / S3 (WP) (R.S) / S1 (PT) (S,FE) Examination November 2024 (2019 Scheme)

Course Code: CST205

Course Name: OBJECT ORIENTED PROGRAMMING USING JAVA

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks Marks

1 State the importance of activity diagram. (3)

2 Explain how byte code helps in achieving platform independence. (3)

3 Define a Constructor? State its properties. (3)

4 State the uses of this keyword. (3)

5 Outline the use of final keyword. (3)

6 Give the difference between checked and unchecked exceptions? (3)

7 List out any three event sources and mention its event type and listeners used in it. (3)

8 Outline the use of synchronized keyword. (3)

9 Explain the difference between AWT and Swing. (3)

10 List any four container and component classes of Swing API? (3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

11 a) Explain various lexical issues in java programming (7)

b) Draw a class diagram for an order processing system (7)

12 a) Name the four integer types in Java and outline the bitwise operators that can be applied to the integer types. (7)

b) Design a use case diagram for a simple online shopping system. Identify the main actors involved and the primary use cases that represent the interactions between the actors and the system. (7)

Module 2

13 a) Differentiate between method Overloading and Overriding with examples. (8)

b) Explain the uses of super keyword with examples. (6)

14 a) Discuss different access specifiers used in java?

b) Explain dynamic method dispatch with the help of an example. (6)

Module 3

(8)

15 a) What is the difference between the Reader /Writer class hierarchy and the Input stream/Output stream class hierarchy? (7)

b) Explain how the following scenario is implemented. Create a package "reversepackage". Add a class Reverse into it and write a method rev () which reverses a number. Import this package in another class outside the package and find the reverse of a number given as command line argument (7)

16 a) What are the exception handling keywords in java?

b) Write a java program to copy the contents of a text file into another text file. (7)

Module 4

(7)

17 a) Outline the concept of adapter classes with the help of an example

b) Explain the Thread Life Cycle with a neat diagram? (8)

18 a) Explain Delegation event model in java with a program example? (6)

b) Explain the different ways of creating a thread using java. (7)

Module 5

(7)

19 a) Outline the various steps involved in connecting java with a database. (8)

b) Write a simple swing program which prints a message on a text box when a button is clicked (6)

20 a) Write a swing program to develop a mini calculator which include addition, subtraction, multiplication and division functionality (8)

b) Write a java program which insert a set of values into a database table 'student' (6)

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Pages: 4

Reg No.:_____ Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S3 (S,FE) (FT/WP) (S1 PT) Examination May 2025 (2019 Scheme)

Course Code: CST205

Course Name: OBJECT ORIENTED PROGRAMMING USING JAVA

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks Marks

1	Object-oriented programming uses classes and objects. What are classes and what are objects? What is the relationship between classes and objects?	(3)
2	Why can't we use the "super" and "this" keywords in a static method in java?	(3)
3	When do we declare a method or class as final ? explain with an example?	(3)
4	List out three differences between abstract classes and interfaces.	(3)
5	What is a finally block? Explain with a suitable Example.	(3)
6	What is Object serialization in Java?	(3)
7	List out three differences between List and Set interfaces in Java.	(3)
8	With an example, explain the advantages of iterating a collection using an iterator.	(3)
9	What is the importance of hashCode() and equals() methods?	(3)
10	Difference between start() and run() method of Thread class?	(3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

11	a)	Whether Class diagrams developed using UML can serve as the functional specification of a system. Justify Your Answer.	(4)
	b)	Explain minimum two salient features of function-oriented design approach and object-oriented design approach	(4)
	c)	Write short Notes on JDK, JRE and JVM	(6)
12	a)	Describe in detail any three Object Oriented Programming principles. Illustrate with suitable examples.	(7)

b) Explain Garbage Collection in JAVA with examples

c) Write a Program to demonstrate the finalize() method in java to show that this method is called just before an object is garbage collected

Module 2

13 a) Create a class Box that uses a parameterized constructor to initialize the dimensions of a box.The dimensions of the Box are width, height, depth. The class should have a method that can return the volume of the box. Create an object of the Box class and test the functionalities.

b) Create a school application with a class called Person. Create name and dateOfBirth as member variables. Create a class called Teacher that inherits from the Person class. The teacher will have additional properties like salary, and the subject that the teacher teaches.Create a class called Student that inherits from Person class. This class will have a member variable called studentId. Create a class called College Student that inherits from Student class. This class will have collegeName, the year in which the student is studying (first/second/third/fourth) etc. Create objects of each of these classes, invoke and test the methods that are available in these classes.

14 a) Design a class that can be used by a healthcare professional to keep track of a patient's vital statistics. The following are the details. (5)

Name of the class - Patient

Member Variables - patientName(String),height(double),width(double)

Member Function - double computeBMI()

The above method should compute the BMI and return the result. The formula for computation of BMI is weight(in kg) ÷ height*height(in metres).

Create an object of the Patient class and check the results.

b) Write a program to create a class named shape. It should contain 2 methods, draw() and erase() that prints "Drawing Shape" and "Erasing Shape" respectively. For this class, create three subclasses, Circle, Triangle and Square and each class should override the parent class functions - draw () and erase (). The draw() method should print "Drawing Circle", "Drawing Triangle" and "Drawing Square" respectively. The erase() method should print "Erasing Circle", "Erasing (9)

Triangle" and "Erasing Square" respectively. Create objects of Circle, Triangle and Square with base class type and explain the polymorphic behaviour of methods

Module 3

15 a) Write a class that implements the Queue interface, as shown below. A queue is a data structure that accepts data and then returns it in the order in which it was received (first-in, first-out order). Items are added to the tail of the queue and removed from the head. (9)

```
public interface Queue {
 public int size(); //Returns number of objects in queue
 public boolean isEmpty(); //Returns true if queue is empty
 //Adds an item to the tail of the queue
 public void enqueue(Object o);
 //Removes and returns the item from the head of the queue
 public Object dequeue(); }
```

Your queue implementation must be accessible and usable from any package. However, any attempt to extend your class should produce a compile-time error.

b) Explain the concept of Serialization and Deserialization in Java with suitable Examples. (5)

16 a) Change the divide method below so that it throws an IllegalArgumentException with an appropriate message if b is zero (3)

```
public static double divide(double a, double b) {
 return a / b; }
```

b) Change the below method so that it catches the IllegalArgumentException thrown by divide method (above question) and prints the message "the divisor is zero" if it is thrown. (3)

```
public static void printQuotient(double a, double b) {
 System.out.println(divide(a, b));
 }
```

c) Write a JAVA program to copy the content of one file to another. mention all the necessary import statements and your program should read the file line by line. (8)

Module 4

17 a) Write a Java program to check whether two given strings are anagram of each other or not. An anagram of a string is another string that contains the same characters, only the order of characters can be different. For example, "abc" and "bac" are an anagram of each other. (8)

b) Create an ArrayList that can store only Strings. Create a printAll method that will print all the elements of the ArrayList using an Iterator. (6)

18 a) Write a Java program to print all permutations of a given string (8)

b) Explain minimum three collection interfaces provided by Java Collection framework (6)

Module 5

19 a) Write a Program which adds JLabel, JButton, JTextField to a JFrame. Your program should display the text field as "Welcome to Java Swing" when clicking on the button. (8)

b) Consider the following employee schema (6)

employee(empid:integer, empname:string, salary:integer, Department:string)

Write a Java Program to create an employee table and to add employee details to it using JDBC

20 a) Design and implement a simple calculator GUI with the following fields (10)

1) two text fields to enter the inputs.

2) a text field to display the output of the operation.

3) addition and subtraction buttons to perform the operation.

4) your program should compute the proper results according to the action performed by the user.

b) Explain the sequence of steps involved in [illegible] (4)

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Pages: 2

Reg No.:_____ Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S3 (R, S) / S1 (PT) (S, FE) Examination December 2023 (2019 Scheme)

Course Code: CST205

Course Name: OBJECT ORIENTED PROGRAMMING USING JAVA

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks Marks

1		Describe the role of JIT compiler.	(3)
2		Why Java is said to be robust?	(3)
3		List the properties of a constructor.	(3)
4		Discuss the use of public, private and protected access specifiers.	(3)
5		Distinguish between checked and unchecked exceptions.	(3)
6		Explain any three character stream classes in Java	(3)
7		Explain the different methods of creating threads in Java.	(3)
8		List any three event listeners and the corresponding source object.	(3)
9		Describe the features of Swing.	(3)
10		Explain about the containers in Swing.	(3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

11	a	Distinguish between function oriented and object oriented design paradigms with an example.	(8)
	b	Draw a UML activity diagram for an order processing system.	(6)

OR

12	a	Explain any four object oriented features of Java with suitable examples.	(8)
	B	Draw a UML class diagram which shows structure and attributes of an ATM with entities Bank, ATM, Debit Card, Customer, Account, ATM Transaction.	(6)

Module 2

13	a	Create a class Person with attributes name, age, address and a method display() to display the details. Create a subclass Student with attributes rollno, mark1, mark2 and mark3. Overide display() method and calculate the	(8)

grade. Create another subclass Faculty with attributes faculty_id, department, basic_pay and DA. Override display() method to calculate total salary as (basic_pay+DA)+70% of (basic_pay+DA). Create instances of Student and Faculty and display the details.

b Explain method overloading with an example. (6)

OR

14 a Explain dynamic method dispatch with an example (8)

b Discuss the uses of final keyword citing examples. (6)

Module 3

15 a Create a package reversepackage. Add a class Reverse in it with a method reverse() to print the reverse of a string without using built-in methods. Create a class outside the package and use this method to reverse a string. (8)

b Explain object stream serialization with sample code. (6)

OR

16 a How do you create user defined exceptions in Java? Write a Java program to find the average of n numbers entered as command line arguments. Raise a user defined exception if the average exceeds 100. (8)

b Explain the use of interface in Java. (6)

Module 4

17 a Write a multithreaded Java program with two threads. One thread generates even numbers from 1 to 100 and another thread generates odd numbers from 1 to 100. Ensure that the threads do not interfere. (8)

b Discuss the methods for String comparison with examples. (6)

OR

18 a Explain how event handling is done in Java. (8)

b How do you access the elements in an ArrayList? Illustrate with an example (6)

Module 5

19 a Explain with sample code the steps in establishing JDBC connectivity. (8)

b Describe the architecture of JDBC with a neat diagram. (6)

OR

20 a Write a GUI program to perform arithmetic operations of two numbers. (8)

b Discuss any three layout managers in Swing. (6)

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Reg No.:_____ Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S3 (S,FE)/S1 (PT)(S) June 2024 (2019 Scheme)/S3 (WP)(R) December 2023 Examination

Course Code: CST205

Course Name: OBJECT ORIENTED PROGRAMMING USING JAVA

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks Marks

1	Why java is said to be platform independent?	(3)
2	Explain any three OOP concepts.	(3)
3	What is a parameterized constructor? Is it possible to create an object for a class A using A obj = new A (), if the class contains only parameterized constructor? Justify	(3)
4	State the uses of super keyword?	(3)
5	Present an outline of Java's checked exceptions defined in java. lang.	(3)
6	What is the need for thread synchronization? How it can be implemented?	(3)
7	Differentiate between interface and abstract class?	(3)
8	What is an event? Explain any three event classes?	(3)
9	Outline the use of setSize(), setVisible() and setTitle() methods when working with frame windows with their signature	(3)
10	Discuss the advantages of Swing over AWT.	(3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

11	(a) Represent a Library management system using a use case diagram	(6)
	(b) Name the four integer types in Java and outline the bitwise operators that can be applied to the integer types.	(8)
12	(a) Explain any six buzzwords using which java can be characterized.	(6)
	(b) Draw the class diagram of a movie reservation system.	(8)

Module 2

13 (a)Differentiate method overloading and method overriding with help of examples

(b) Write a java program to find the sum of two complex numbers. Use suitable constructors and methods.

14 (a)Write a java program to find the product of two matrices?

(b)Define Constructor? State its properties. Explain its uses with examples

Module 3

15 (a)Explain the use of try, catch, throw, throws and finally in detail?

(b) Write a Java program to read a string from a file and store it in another file

16 (a) Write a java program that catches NegativeException. This is caused when a negative number is entered by user?

(b)What are the steps involved in creating a user defined package with example?

Module 4

17 (a) Explain List and Collections interfaces. Mention the exceptions thrown by its methods

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Module 2

13 (a)Differentiate method overloading and method overriding with the help of examples (7)

(b) Write a java program to find the sum of two complex numbers. Use suitable constructors and methods. (7)

14 (a)Write a java program to find the product of two matrices? (7)

(b)Define Constructor? State its properties. Explain its uses with examples. (7)

Module 3

15 (a)Explain the use of try, catch, throw, throws and finally in detail? (7)

(b) Write a Java program to read a string from a file and store it in another file (7)

16 (a) Write a java program that catches NegativeException. This is caused when a negative number is entered by user? (6)

(b)What are the steps involved in creating a user defined package with example? (8)

Module 4

17 (a) Explain List and Collections interfaces. Mention the exceptions thrown by its methods. (6)

(b)Write a program to count the total no. of chars, words, lines, alphabets, digits, white spaces in a given string (8)

18 (a)What is meant by event delegation model? (6)

(b)Write a program that creates two threads. Fist thread prints the numbers from 1 to 100 and the other thread prints the numbers from 100 to 1. Execute the two threads in parallel (8)

Module 5

19 (a)What are the steps followed to establish a connection with JDBC driver and a JAVA application (7)

(b)Write a Swing program to create a mini calculator. (7)

20 (a)List and explain different types of Layout managers with suitable examples. (7)

(b)Write a java GUI Program to calculate the factorial of a number. (7)

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Pages: 3

Reg No.:_____ Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme)

Course Code: CST205

Course Name: OBJECT ORIENTED PROGRAMMING USING JAVA

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks Marks

1		How is platform independence achieved in Java?	(3)
2		Explain how garbage collection is done in Java.	(3)
3		Explain the use of static variable with the help of an example.	(3)
4		Can final modifier be used with an abstract class. Justify your answer.	(3)
5		Differentiate between the usage of keywords throw and throws.	(3)
6		Explain the significance of CLASSPATH environment variable in Java.	(3)
7		List any three event sources and their corresponding event types and listeners used.	(3)
8		Illustrate the creation of arraylist with the help of a sample program.	(3)
9		Compare Swing API and AWT API.	(3)
10		What are layout managers? List any twolayout managers.	(3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

11	a	Consider the problem of a Service Station which provides three types of services to its customers: refuelling, vehicle maintenance and parking. Customer can pay using cash, card or cheque. The pricing for vehicle maintenance depends on the cost of parts and labour. Parking areas are rented according to weekly and monthly rates. Construct an UML class diagram for the above problem by identifyingatleast six entities in the system which can be represented using classes and show the relationship between them.	(10)
	b	Describe programming structure of Java that deals with the organization of Java code.	(4)
12	a	Differentiate between the two main approaches of software design.	(7)

b Construct a UML Activity diagram for an online Hotel Reservation System, which shows the flow of activities for booking rooms at a hotel. (7)

Module 2

13 a Write a Java program by creating a 'Student' class having the following data members: rollNumber, name, mathMarks, phyMarks, chemMarks and methods getRequiredDetails() – to get required input and displayAverage() – to calculate average marks and display it. In class 'Implement' create an object of the Student class and get the required details from user and display the average marks of that student. (7)

b Write a java program that illustrates how 'this' keyword can be used to resolve the ambiguity between formal parameters and instance variables. (7)

14 a Explain the concept of method overloading with the help of a program. (7)

b What is inheritance? Illustrate hierarchical inheritance using a sample program. (7)

Module 3

15 a Write a program to read the first n characters in a file where n is given by the user. The characters read from file has to be reversed and displayed on screen. Built in methods can be used in the program. (7)

b Explain the role of access modifiers when packages are used in Java. (7)

16 a Create a user defined exception 'InvalidAgeException'. Write a Java program that takes age as a Command Line Argument. Raise the Exception 'InvalidAgeException' if age is less than 18. (7)

b Explain the concept of Serialization and demonstrate how an object can be serialized with a sample program. (7)

Module 4

17 a Illustrate the event handling mechanism in Java using the Delegation Event Model with the help of a diagram. (8)

b Illustrate the usage of the following methods related to String with appropriate sample code. (6)

(i) find() (ii)substring() (iii) replace()

18 a What is multithreading? Write a multithreaded Java program that demonstrates the working of wait() and notify() methods. (7)

	b	Explain how ActionEvent class and FocusEvent class is used with emphasis on the methods and constants provided by the given classes.	(7)

Module 5

19	a	Write a Java program that uses two textfields and a button. The first textfield accepts temperature in Celsius. When the 'Convert' button is clicked the second textfield displays the temperature in Fahrenheit. Use appropriate Swing components and event handling techniques. F=(C*9/5)+32	(9)
	b	Describe the two different ways to create frames using Swing package with appropriate examples.	(5)
20	a	Discuss the Model View Controller (MVC) Architecture using a diagram. Also list out the advantages of writing programs based on MVC Architecture.	(7)
	b	Explain the various steps for connecting to database using JDBC API, using a sample program.	(7)

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Reg No.:_____ Name:_____

LIBRARY

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: CST205

Course Name: OBJECT ORIENTED PROGRAMMING USING JAVA

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks Marks

1 What is Just-In-Time compiler? (3)

2 Why Java is said to be a secure programming language? (3)

3 Does Java support multiple inheritance? Justify your answer. (3)

4 Why is the 'main' method in Java qualified as public, static, and void? (3)

5 Explain any three Byte Stream classes in Java. (3)

6 What are Checked Exceptions? Give an example. (3)

7 List any six Event Listener interfaces. (3)

8 Explain any three special string operations in Java (3)

9 Summarize any three features of Swing API. (3)

10 Differentiate between Components and Containers in Swing. (3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

11 a) Explain Lexical issues in Java with examples. (6)

b) Differentiate between function oriented and object orientedsoftware design approaches using a suitable example. (8)

12 a) Construct a UML Class diagram for Online Movie Ticket Booking System. The various entities involved in the system are Admin, Registered User, Visitor / Guest User, Movie, Book Ticket, Make Payment. (8)

b) Construct a UML Activity diagram for Food Ordering System, which shows the flows between the activity of Order, Delivery, Food Item, Category, Payment. (6)

Module 2

13 a) Explain different data types in Java. Give examples. (8)

b) Write a Java program to reverse bits of a given integer. (6)

14 a) Demonstrate how objects are passed as function parameters with a suitable example. (8)

b) Write a Java program to find the frequency (count the occurrence) of each element in an integer array. (6)

Module 3

15 a) Develop a Java package named '*evenpackage*', with a class *Even* containing a static method that check whether a number is even or not, and returns that (8)

information. Import this package in another class and use to check a number is even or not.

b) Differentiate between *try-catch*, *throw* and *throws* keywords. (6

16 a) Write a Java program that reads a binaryfile and write to another file. (8

b) Write Java code that reads a character file and prints the contents of file on the display, with a line number before each line. (6

Module 4

17 a) Write a Java program to find the duplicate characters in a string. (8)

b) What are the uses of *synchronized* keyword in Java? Explain with examples. (6)

18 a) Write a Java program that creates three threads. First thread generates a random positive number (>1) every 1 second. If the number is even, the second thread prints all even numbers between 1 and the generated number. If the number is odd, the third thread will print all odd numbers between 1 and the generated number. (10)

b) Differentiate between Collection Interface and Collections Class. (4)

Module 5

19 a) How do you establish connection between a Java program and database? Explain the steps with sample code. (10)

b) Write java code to demonstrate the execution of select and delete queries using JDBC (4)

20 a) How events are handled in java Swing? (4)

b) Write a Java program using Swing to create a frame having three text fields, threelabels and a button. The interface has to accept a number in the first text field. While clicking the button, the second and third textfields have to display the previous number and next number respectively, of the accepted input number. (10)

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